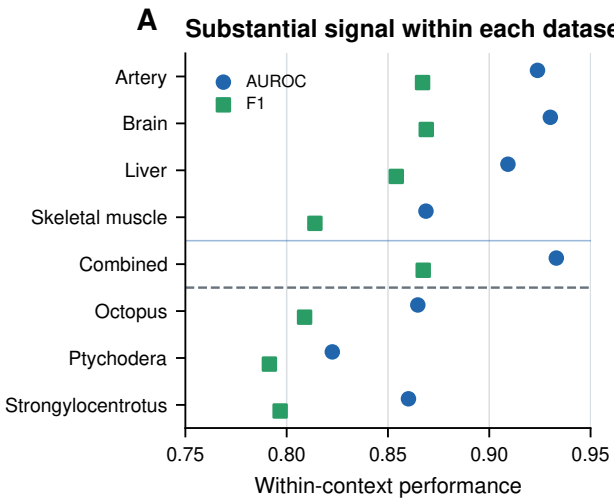
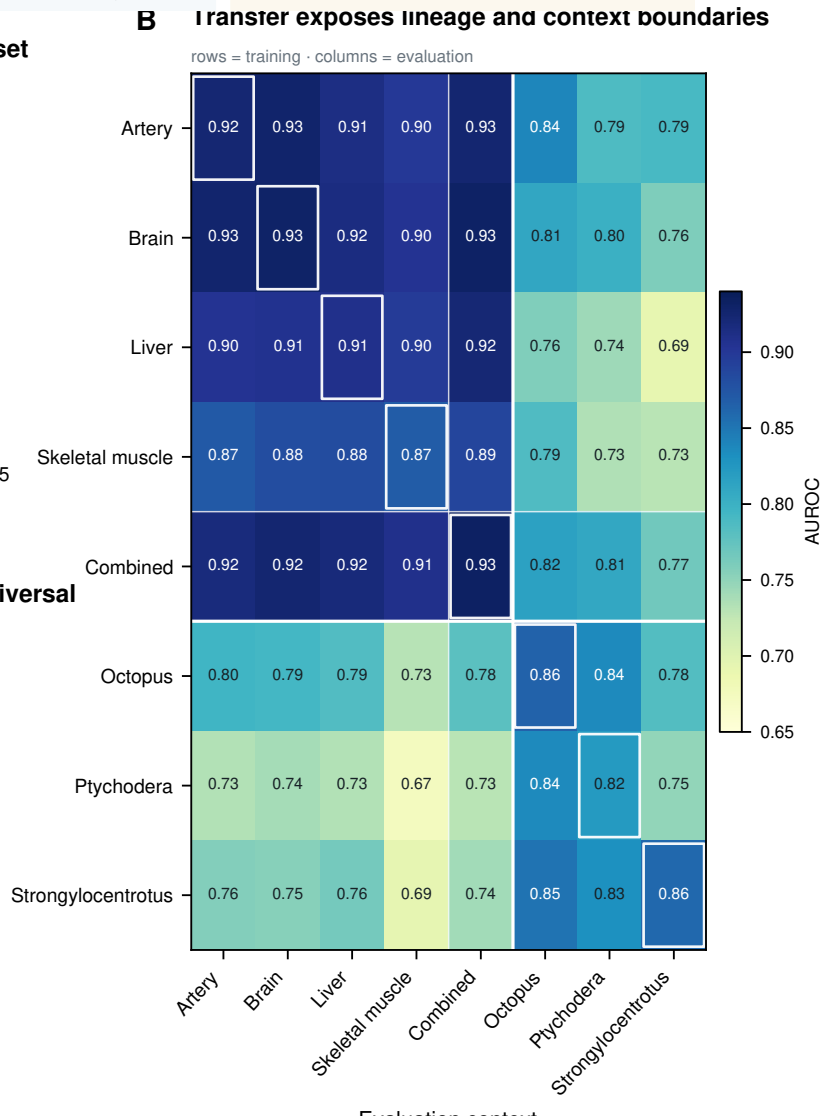
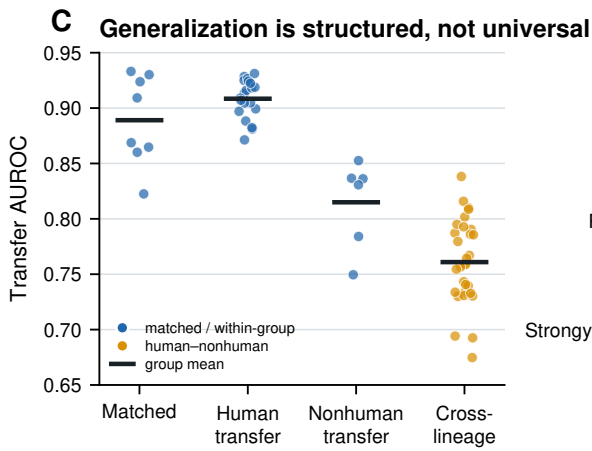

HUMAN ENDPOINTS
Artery
Brain
Liver
Skeletal muscle
Combined (pooled)
NONHUMAN DATASETS
Octopus
Ptychodera
Strongylocentrotus


A
Substantial signal within each dataset
Artery
Brain
Liver
Skeletal muscle
Combined
Octopus
Ptychodera
Strongylocentrotus
AUROC
F1
0.75
0.80
0.85
0.90
0.95
Within-context performance


B
Transfer exposes lineage and context boundaries
rows = training · columns = evaluation
Artery 0.92 0.93 0.91 0.90 0.93 0.84 0.79 0.79
Brain 0.93 0.93 0.92 0.90 0.93 0.81 0.80 0.76
Liver 0.90 0.91 0.91 0.90 0.92 0.76 0.74 0.69
Skeletal muscle 0.87 0.88 0.88 0.87 0.89 0.79 0.73 0.73
Combined 0.92 0.92 0.92 0.91 0.93 0.82 0.81 0.77
Octopus 0.80 0.79 0.79 0.73 0.78 0.86 0.84 0.78
Ptychodera 0.73 0.74 0.73 0.67 0.73 0.84 0.82 0.75
Strongylocentrotus 0.76 0.75 0.76 0.69 0.74 0.85 0.83 0.86
Artery
Brain
Liver
Skeletal muscle
Combined
Octopus
Ptychodera
Strongylocentrotus
Evaluation context
AUROC
0.90
0.85
0.80
0.75
0.70
0.65


C
Generalization is structured, not universal
Transfer AUROC
0.95
0.90
0.85
0.80
0.75
0.70
0.65
matched / within-group
human–nonhuman
group mean
Matched
Human transfer
Nonhuman transfer
Cross-lineage